THE COMPLETE INTERVIEW GUIDE

Master the *RAG* Interview

The essential components, concepts, and terminology you must understand deeply to clear a Retrieval Augmented Generation interview.

Pipeline

Embeddings

Retrieval

Reranking

Evaluation

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The mental *model*

RAG stands for Retrieval Augmented Generation. Every interview question maps to one stage of a single pipeline.

1. **Ingest and chunk.** Split source documents into retrievable pieces.
2. **Embed and index.** Turn chunks into vectors and store them for fast search.
3. **Retrieve.** Fetch the most relevant chunks for the query.
4. **Augment and generate.** Place that context into the prompt and answer.
5. **Evaluate.** Measure whether the answer is grounded and relevant.

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Stage 1

Ingestion and *chunking*

Retrieval quality is decided here, long before the model sees anything.

- **Chunking.** Splitting documents into smaller units for indexing and retrieval.
- **Chunk size and overlap.** The balance between full context and sharp precision.
- **Semantic chunking.** Splitting on meaning and structure, not a fixed character count.
- **Metadata.** Tags such as source, date, and section, used later to filter results.
- **Document loaders.** The parsers that turn PDFs, HTML, and tables into clean text.

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Embeddings

The layer that lets a machine compare meaning, rather than exact words.

- **Embedding.** A dense vector that captures the meaning of a piece of text.
- **Dense vs sparse.** Dense captures semantics, sparse captures exact keywords.
- **Similarity metric.** How closeness is measured, most often cosine similarity.
- **Dimensionality.** The size of the vector, which affects both cost and accuracy.
- **Embedding model.** The model that produces the vectors, chosen to match your domain.

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Vector stores and *indexing*

Where the vectors live, and how they are searched at scale.

- **Vector database.** A store built to search millions of vectors quickly.
- **Approximate nearest neighbour.** Trading a little accuracy for large gains in speed.
- **HNSW and IVF.** The index structures that make that search fast.
- **Metadata filtering.** Narrowing the search by tags, before or after retrieval.
- **Recall.** The share of truly relevant chunks the search actually returns.

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Retrieval

The step that decides what the model is even allowed to see.

- **Top k.** The number of chunks returned for each query.
- **Semantic search.** Retrieval by meaning, through vector similarity.
- **Hybrid search.** Combining dense vectors with keyword search such as BM25.
- **Query rewriting.** Reshaping a vague query into one that retrieves better.
- **Multi query and HyDE.** Generating extra queries or a draft answer to widen recall.

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Stage 3

Re-ranking

A second pass that fixes what fast retrieval gets wrong.

- **Reranker.** A model that reorders retrieved chunks by true relevance.
- **Cross encoder.** Scores the query and chunk together for higher precision.
- **Two stage retrieval.** Retrieve widely and cheaply, then rerank the top results.
- **Relevance score.** The signal used to keep only the strongest context.

Rule of thumb. Rerank when precision matters more than raw speed.

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Generation and *grounding*

Turning retrieved context into an answer the user can trust.

- **Augmentation.** Placing retrieved context into the prompt before generation.
- **Grounding.** Tying every claim in the answer to the retrieved sources.
- **Context window.** The token budget that limits how much context fits.
- **Hallucination.** A fluent answer that the sources do not actually support.
- **Citations.** Linking each claim back to the chunk it came from.

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Evaluation

How you prove the system works. This is where senior candidates separate themselves.

- **Faithfulness.** Whether the answer stays true to the retrieved context.
- **Answer relevance.** Whether the answer actually addresses the question.
- **Context precision.** Whether the retrieved chunks were on topic.
- **Context recall.** Whether all the needed information was retrieved.
- **RAGAS.** A common framework for measuring these scores.

Advanced *patterns*

The terms that signal you have gone beyond the basic pipeline.

- **Agentic RAG.** An agent decides when to retrieve, and what to retrieve.
- **GraphRAG.** Retrieval over a knowledge graph for multi hop questions.
- **Self RAG.** The model critiques its answer and retrieves again to improve it.
- **Contextual retrieval.** Adding context to each chunk before it is embedded.
- **Reranking cascades.** Layering cheap and expensive rankers in sequence.

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The depth they *probe*

Interviewers rarely want a definition. They want the trade off behind it.

- **Chunking.** Small chunks are precise but lose context. Large chunks do the reverse.
- **Cosine similarity.** Preferred because it compares direction, not magnitude.
- **Recall vs precision.** Retrieve too little and you miss. Too much and you add noise.
- **Why rerank.** Vector search is fast but approximate. Reranking restores precision.
- **Why measure retrieval alone.** Most RAG failures are retrieval failures, not model failures.

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Answer with *depth*

Strong answers connect a term to a decision, and a decision to a trade off.

1. Name the term, then the trade off it manages.
2. Tie every choice back to retrieval quality or cost.
3. Separate retrieval failures from generation failures.
4. Always say how you would measure it.
5. Ground it in a system you have actually built.

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If you remember *one thing*

RAG is not a single model. It is a pipeline, and each stage is a place it can quietly fail.

- **Master the pipeline,** and every term finds its place.
- **Retrieval quality** decides answer quality.
- **Precision and recall** are the two forces you always balance.
- **Evaluation is not optional.** It is how you prove the system works.

Do this, and you will explain RAG with the confidence of someone who has built it.

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Save this for your RAG interview prep.

If it helped you see RAG as one clear pipeline, share it with someone preparing for the same interview.

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